

GANESH GUPTA

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

Experience

Oracle

Jun 2025 – Present

Software Engineer, AI Platform

- Built a multi-agent system in Python using the OpenAI Agents SDK, LangGraph, and MCP tool-calling, with prompt engineering and agentic reasoning loops handling millions of daily transactions at 99%+ uptime.
- Designed stateful agent recovery and context management via Redis checkpointing, so interrupted agent workflows resume from the last committed step across distributed, fault-tolerant execution.
- Deployed a RAG pipeline on AWS SageMaker for real-time decision support at sub-200ms p99 latency, integrating retrieval-augmented generation into production agentic workflows.

QureAI (AI Startup)

Mar 2026 – Present

ML Infrastructure Engineer

- Migrated synchronous PyTorch-based model serving to async vLLM and TensorRT-LLM with autoscaling, cutting p99 inference latency 420ms to 140ms across 50K+ daily requests.
- Integrated AWS Bedrock-hosted foundation models into a production voice agent, extending agentic workflows from text to voice-based interaction as an early engineer at this AI startup.

Nomura Research Institute Financial Technologies

Sep 2019 – Aug 2023

Software Engineer

- Architected and operated 25+ production C++ and SQL-backed microservices in distributed systems for 1M+ users, sustaining 99.9% uptime over 4 years on AWS (EC2, RDS, SQS).
- Drove a redesign of nightly data aggregation, cutting processing from 8 minutes to 90 seconds via indexed views and batch partitioning.
- Led standardization of failure-handling contracts (retry, timeout, propagation semantics) across 25+ services in partnership with cross-functional teams through code reviews, reducing production escalations 60% in one quarter.

Projects

RootCause AI | github.com/ganeshhgupta/rootcause-ai

- LangGraph multi-agent system diagnosing production incidents via tool-use chains; routes alerts to specialized sub-agents and synthesizes evidence-backed root cause reports, reducing mean time to resolution.

LangFetch | tinyurl.com/LangFetch

- LangGraph agent with Pinecone vector retrieval and hybrid re-ranking decomposes multi-table questions into dependency-ordered SQL subqueries; deployed as an MCP-compatible server for enterprise data queries.

Education

University of Texas at Arlington

Aug 2023 – May 2025

MS, Computer Science, GPA 4.0/4.0

MCKV Institute of Engineering

Jul 2016 – Jul 2020

BTech, Computer Science

Skills

Languages: Python, TypeScript, SQL, Java, C++

Agentic AI & LLM: Agentic AI, Multi-Agent Systems, Voice Agents, LangChain, LangGraph, OpenAI Agents SDK, Anthropic Claude, AWS Bedrock, MCP, RAG, Prompt Engineering, Tool/Function Calling

Inference & ML: PyTorch, vLLM, TensorRT-LLM, Vector Databases (Pinecone, FAISS), Model Evaluation

Backend, Cloud & MLOps: FastAPI, Flask, Node.js, REST APIs, Microservices, AWS (Lambda, EC2, RDS, SQS, SageMaker, Bedrock), Azure, GCP (Cloud Run, Vertex AI), Docker, Kubernetes, CI/CD

Certifications

AWS Certified Developer Associate | GCP Professional MLOps Engineer